

Pizza Sales Analysis

Using SQL

Data Analytics Portfolio Project — transforming raw sales data into actionable business insights through SQL.

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Data Analytics

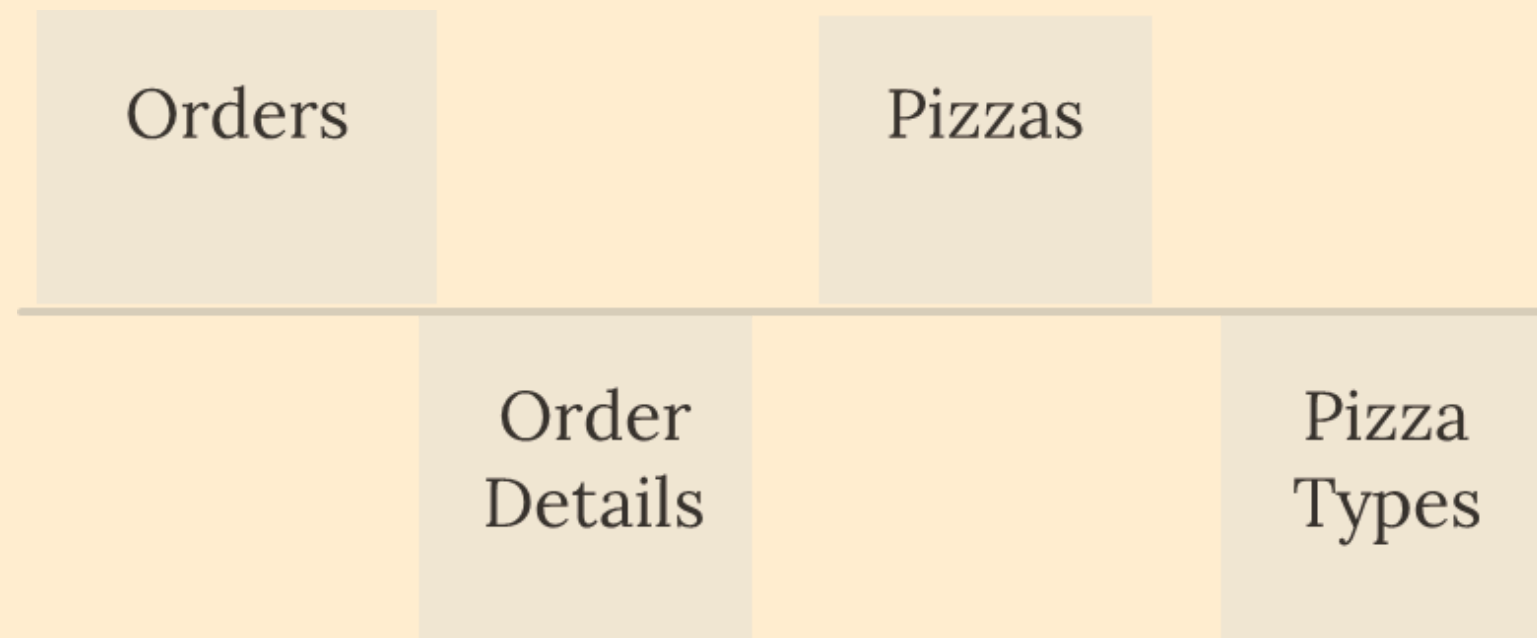
SQL Portfolio



Project Objective

Analyze pizza sales data using SQL to uncover trends, revenue patterns, and customer preferences – demonstrating core analytical skills on a real-world dataset.

Dataset Tables



Skills Demonstrated

→ SQL Joins & Aggregate Functions

→ GROUP BY & CTEs

→ Window Functions

→ Business Intelligence & Reporting

SQL Analysis Breakdown



Basic Analysis

- Total orders & revenue
- Highest-priced pizza
- Most common pizza size
- Top 5 most ordered pizzas

Intermediate Analysis

- Category-wise quantity sold
- Orders by hour of day
- Category distribution
- Average daily orders
- Top revenue-generating pizzas

Advanced Analysis

- Revenue contribution %
- Cumulative revenue over time
- Category-wise top revenue pizzas

Retrieve The Total Number of Orders Placed.

```
SELECT  
    COUNT(order_id) AS total_orders  
FROM  
    orders;
```

	total_orders
▶	21350

List the top 5 most ordered pizza types along with their quantities



```
SELECT
    pizza_types.name, SUM(orders_details.quantity) AS Quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY Quantity DESC
LIMIT 5;
```

Result Grid			Filter Rows:
	name	Quantity	
▶	The Classic Deluxe Pizza	2453	
	The Barbecue Chicken Pizza	2432	
	The Hawaiian Pizza	2422	
	The Pepperoni Pizza	2418	
	The Thai Chicken Pizza	2371	

Determine the distribution of orders by hour of the day

```
SELECT
    HOUR(order_time) AS hour, COUNT(order_id) AS order_count
FROM
    orders
GROUP BY HOUR(order_time);
```

Result Grid		
	hour	order_count
▶	11	1231
	12	2520
	13	2455
	14	1472
	15	1468
	16	1920
	17	2336
	18	2399
	19	2009
	20	1642
	21	1198
	22	663
	23	28
	10	8
	9	1

Determine the top 3 most ordered pizza types based on revenue for each pizza category.



```
( select category , name , revenue , rank() over (partition by category order by revenue desc )
from
(SELECT
    pt.category,
    pt.name,
    SUM(od.quantity * p.price) AS revenue
FROM pizza_types pt
JOIN pizzas p
    ON pt.pizza_type_id = p.pizza_type_id
JOIN orders_details od
    ON p.pizza_id = od.pizza_id
GROUP BY pt.category, pt. name
) as a) as b
where rn <= 3;
```

Result Grid			Filter Rows:
	name	revenue	
▶	The Thai Chicken Pizza	43434.25	
	The Barbecue Chicken Pizza	42768	
	The California Chicken Pizza	41409.5	
	The Classic Deluxe Pizza	38180.5	
	The Hawaiian Pizza	32273.25	
	The Pepperoni Pizza	30161.75	
	The Spicy Italian Pizza	34831.25	
	The Italian Supreme Pizza	33476.75	
	The Sicilian Pizza	30940.5	
	The Four Cheese Pizza	32265.7000000	
	The Mexicana Pizza	26780.75	
	The Five Cheese Pizza	26066.5	

Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT
  pt.category,
  ROUND(
    SUM(od.quantity * p.price) * 100.0 /
    (SELECT SUM(od2.quantity * p2.price)
     FROM orders_details od2
     JOIN pizzas p2
       ON p2.pizza_id = od2.pizza_id
    ),
    2) AS revenue_percentage
FROM pizza_types pt
JOIN pizzas p
  ON pt.pizza_type_id = p.pizza_type_id
JOIN orders_details od
  ON p.pizza_id = od.pizza_id
GROUP BY pt.category
ORDER BY revenue_percentage DESC;
```

Result Grid			Filter Rows:
	category	revenue_percentage	
▶	Classic	26.91	
	Supreme	25.46	
	Chicken	23.96	
	Veggie	23.68	

Conclusion

This project demonstrated how SQL transforms raw sales data into actionable business intelligence — from basic aggregations to advanced window functions and revenue analysis.

 **SQL Query Writing**

 **Joins & Relationships**

 **Data Cleaning**

 **Business Intelligence**

 **Data Cleaning**

 **Window Functions**

Thank you! — Connect with Akshat Nigam on LinkedIn to explore more data analytics projects.